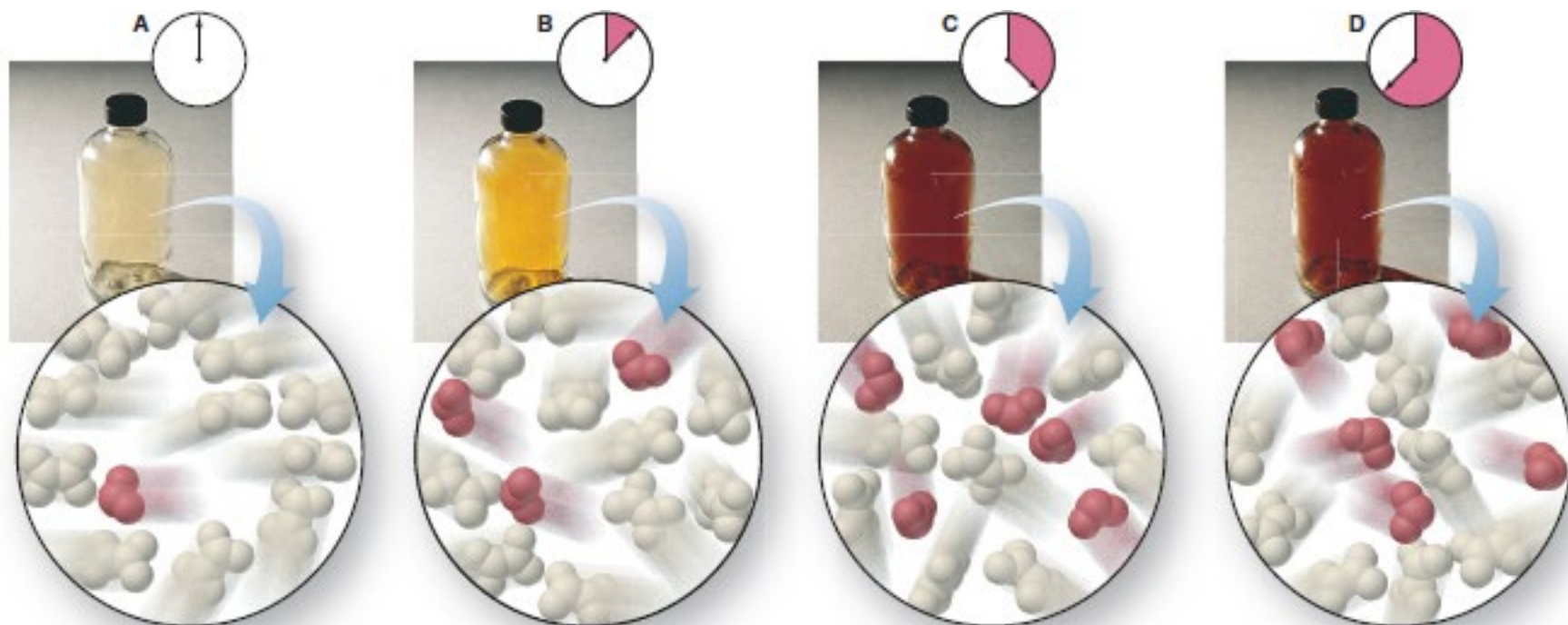


Chapter 17: Kinetics Content Lecture



KINETICS

Central focus:
The rates of chemical reactions
(How fast does the reaction proceed)

Chapter 17: Kinetics– LEARNING OBJECTIVES

By the end of this chapter, you should be able to:

- Differentiate between reaction rates
- Explain and express reaction rate and relate the effects of concentration, physical state, and temperature, on reaction rate
- Differentiate between the average rate and the instantaneous rate
- Explain the experimental basis of the rate law, and describe the information needed to determine it: initial rate data, reaction orders, and rate constant.
- Determine reaction orders
- Calculate the rate constant (k) and its units.
- Calculate change in concentrations over the course of a reaction and determine the half-life of a reaction
- Describe activation energy and the effect of temperature on the rate constant (Arrhenius equation)
- Describe the transition state theory
- Use reaction energy diagrams to depict the energy changes during a reaction.
- Describe the effect of a catalyst on the reaction mechanism

Kinetics

Big Picture for Chemistry

Thermodynamics answers about equilibrium and if a reaction *is possible* under given conditions (spontaneous versus non spontaneous reactions)

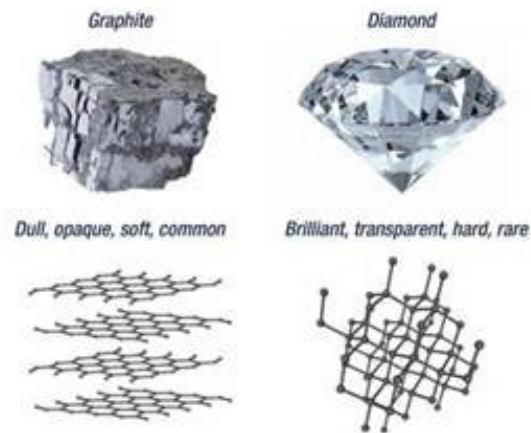
Chemical kinetics measures the **rate** at which a reaction proceeds (how fast will it happen?)

Thermodynamics predicts that diamond would spontaneously convert to graphite ...but the kinetics are very slow

Concentration
Change



Graphite vs Diamond



Chapter 17: Kinetics

17.1-17.2: Reaction Rates

17.3: Rate Law/Reaction Orders/Rate Constant

17.4: Integrated Rate Laws

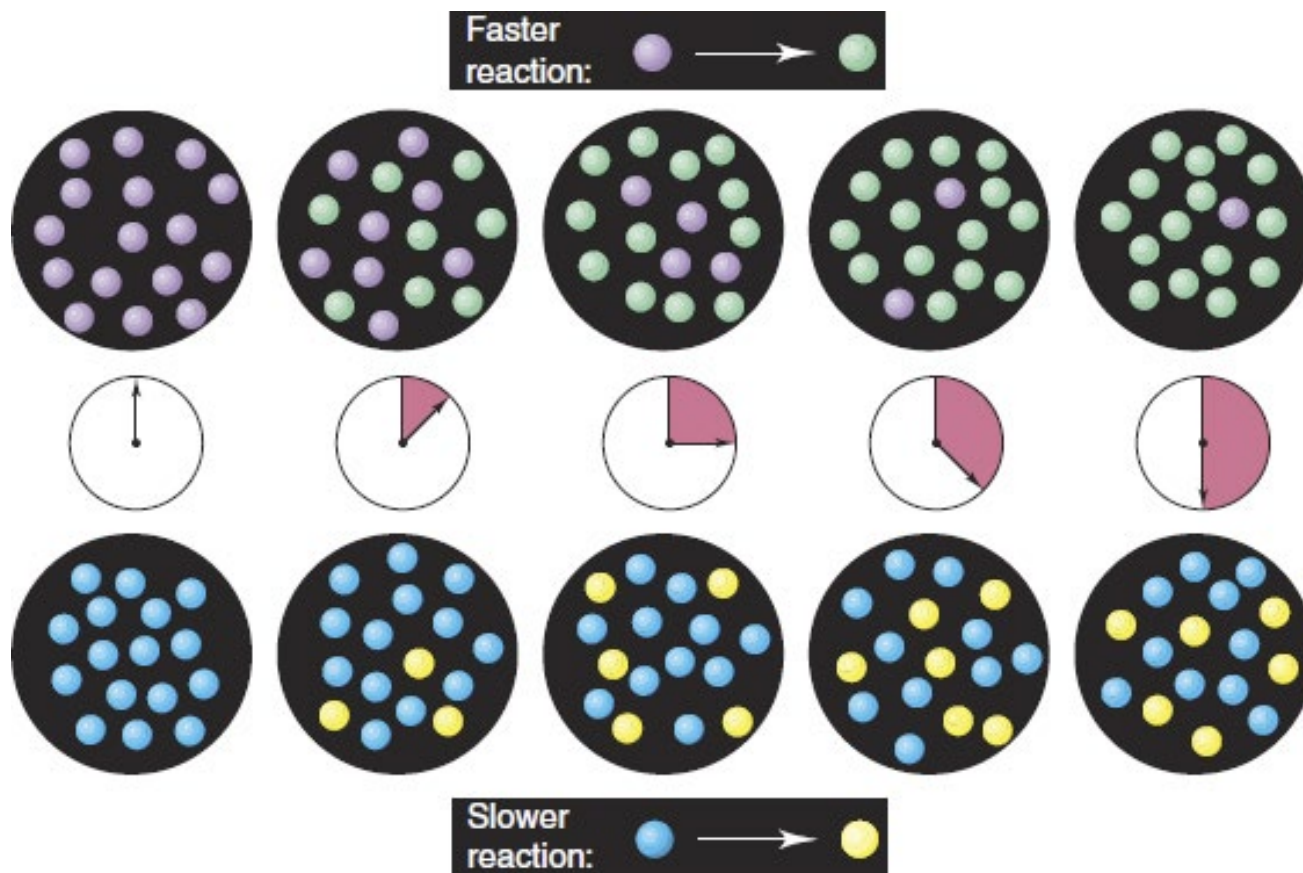
17.5: Collision and Transition State Theory

17.6: Reaction Mechanisms

17.7: Catalysis

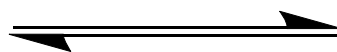
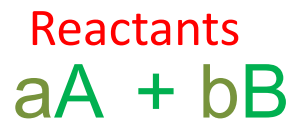
Kinetics

Chemical reaction as function of time



Expressing Reaction Rates

For any reaction:



$$\frac{\Delta[X]}{\Delta t} = \frac{[X]_2 - [X]_1}{t_2 - t_1}$$

$$\text{Units } M s^{-1} = mol L^{-1} s^{-1}$$

Reaction rate can be expressed as: Rate of disappearance of reactants

$$= -\frac{1}{a} \frac{\Delta[A]}{\Delta t} = -\frac{1}{b} \frac{\Delta[B]}{\Delta t}$$

Note negative sign!

Reaction rate can be expressed as: Rate of appearance of products

$$= \frac{1}{c} \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \frac{\Delta[D]}{\Delta t}$$

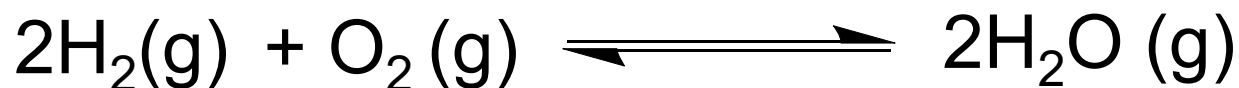
Note positive sign!

Note Normalization by stoichiometric coefficients to get equivalent rates

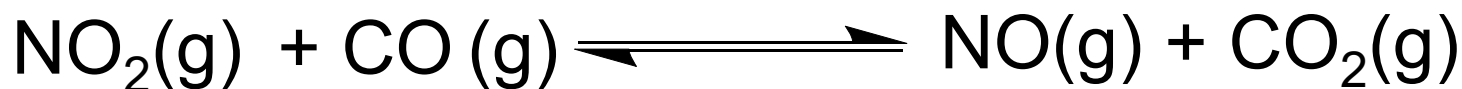
$$\text{rate} = -\frac{1}{a} \frac{\Delta[A]}{\Delta t} = -\frac{1}{b} \frac{\Delta[B]}{\Delta t} = \frac{1}{c} \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \frac{\Delta[D]}{\Delta t}$$

Expressing Reaction Rates

Express the rate of the following reaction in terms of reactants and products:



$$\text{Rate} = -\frac{1}{2} \frac{\Delta[\text{H}_2]}{\Delta t} = -\frac{\Delta[\text{O}_2]}{\Delta t} = +\frac{1}{2} \frac{\Delta[\text{H}_2\text{O}]}{\Delta t}$$



$$\text{Rate} = -\frac{\Delta[\text{NO}_2]}{\Delta t} = -\frac{\Delta[\text{CO}]}{\Delta t} = +\frac{\Delta[\text{NO}]}{\Delta t} = +\frac{\Delta[\text{CO}_2]}{\Delta t}$$

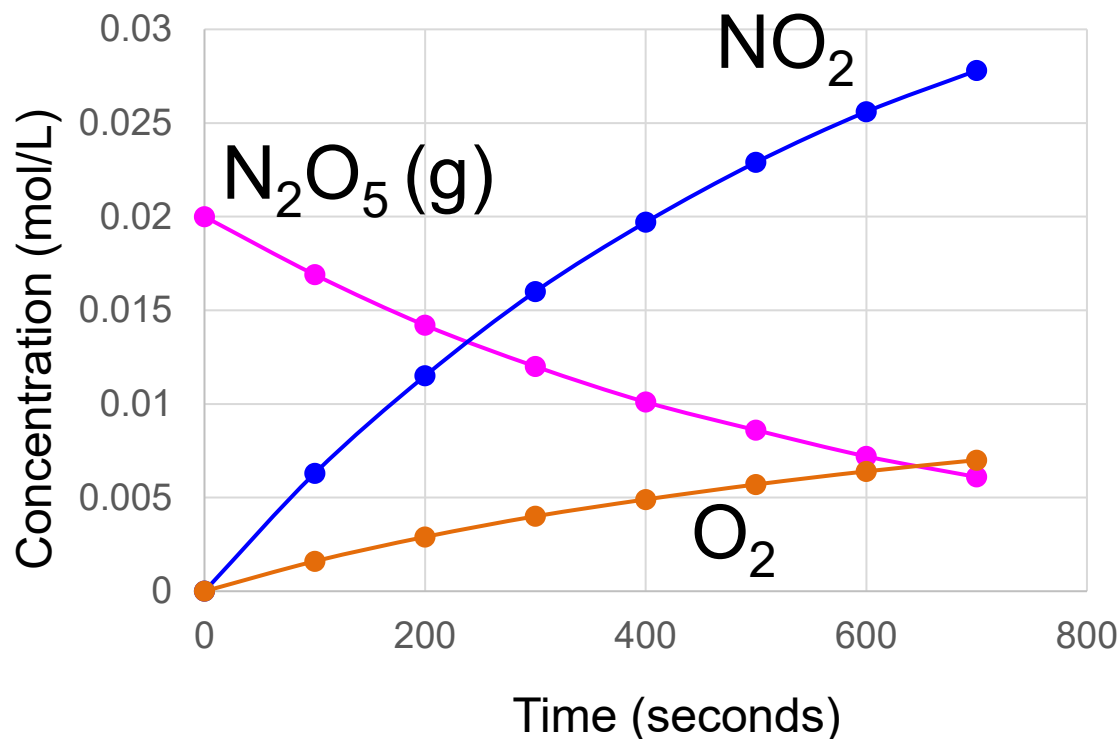


$$\text{Rate} = -\frac{\Delta[\text{N}_2]}{\Delta t} = -\frac{1}{3} \frac{\Delta[\text{H}_2]}{\Delta t} = +\frac{1}{2} \frac{\Delta[\text{NH}_3]}{\Delta t}$$

Expressing Reaction Rates



Measure the concentration of reactants or products as a function of time

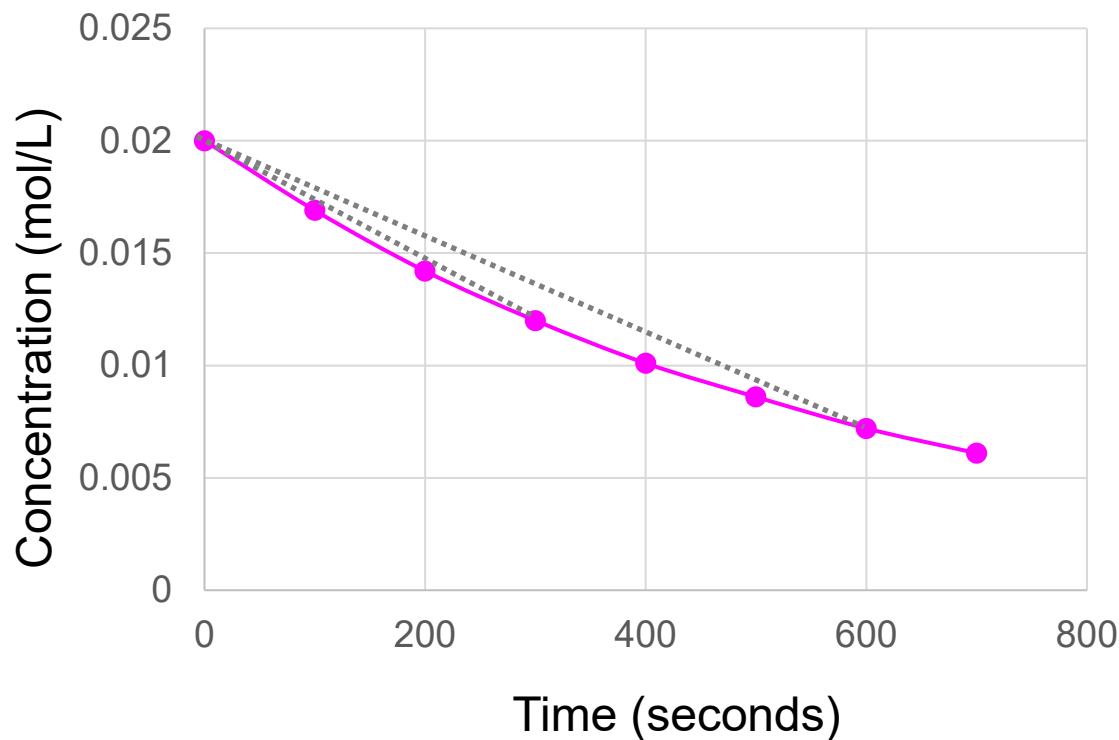


Time (s)	N_2O_5 (mol/L)	NO_2 (mol/L)	O_2 (mol/L)
0	0.02	0	0
100	0.0169	0.0063	0.0016
200	0.0142	0.0115	0.0029
300	0.012	0.016	0.004
400	0.0101	0.0197	0.0049
500	0.0086	0.0229	0.0057
600	0.0072	0.0256	0.0064
700	0.0061	0.0278	0.007

Expressing Reaction Rates



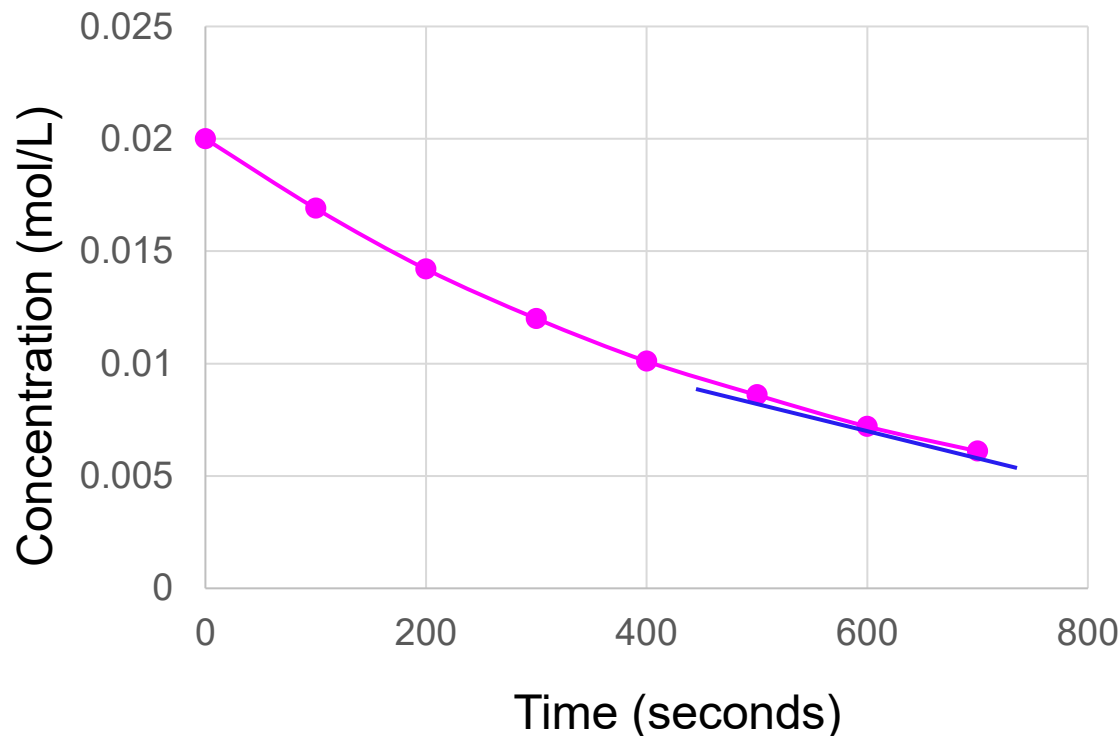
$$\text{Avg. Rate} = \frac{\Delta c}{\Delta t} = \frac{c_2 - c_1}{t_2 - t_1}$$



Average Rate:
depends upon the time
interval chosen.
Equal to the slope of
the line joining the
time interval

Expressing Reaction Rates

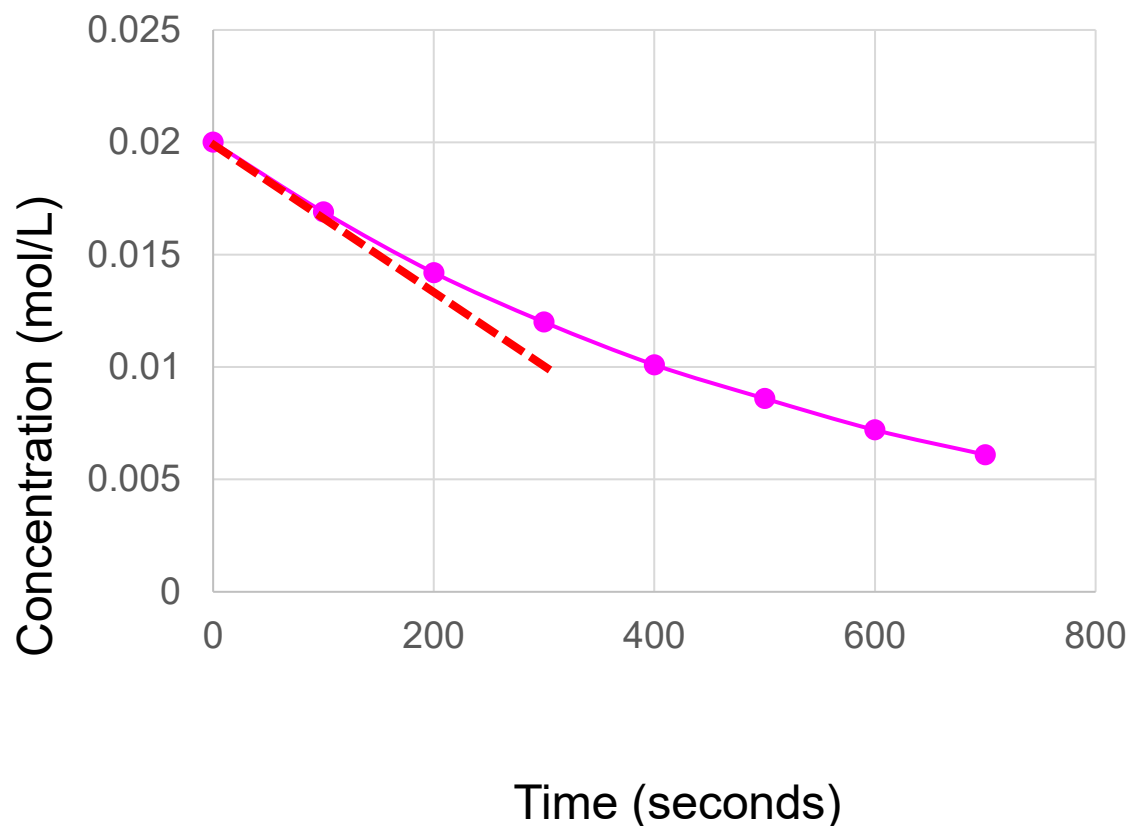
This plot shows the decrease in the concentration of a substance with time



Instantaneous Rate:
The time interval chosen is very short (Δt approaches 0), the **rate is the slope of the line tangent to the curve at time t .**

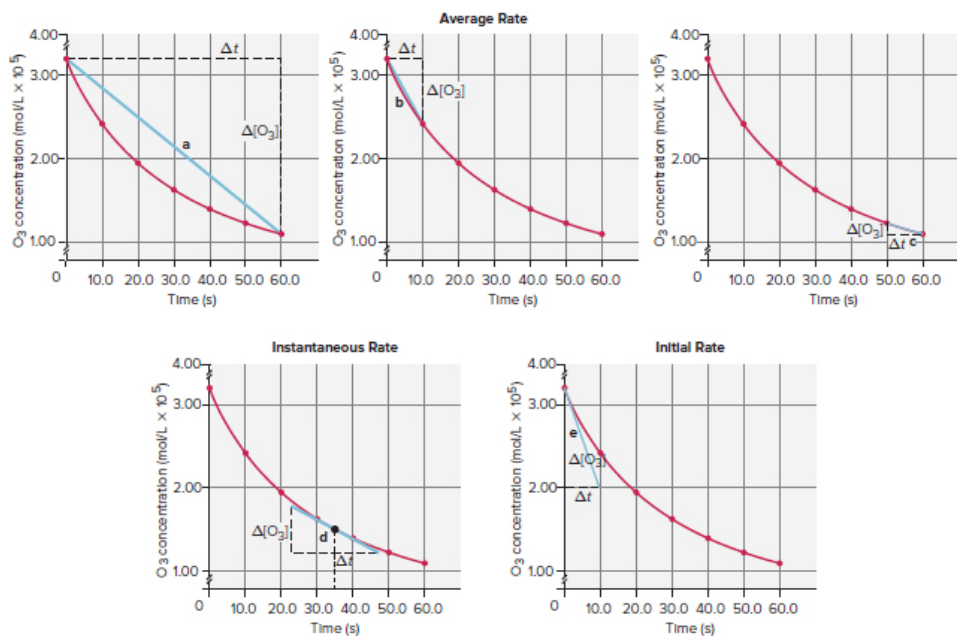
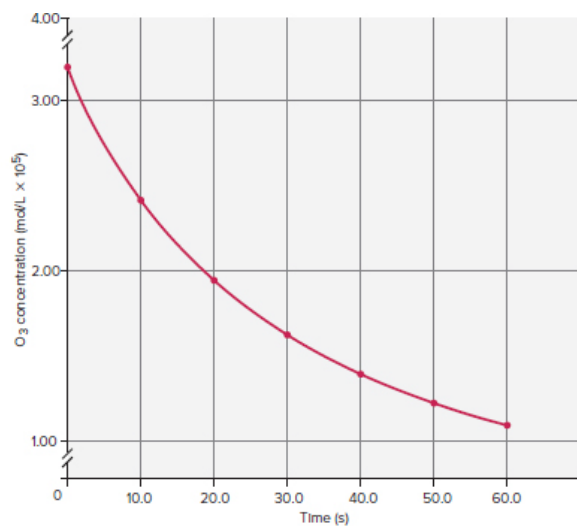
Expressing Reaction Rates

We usually consider the initial rate of the reactants before significant product has formed



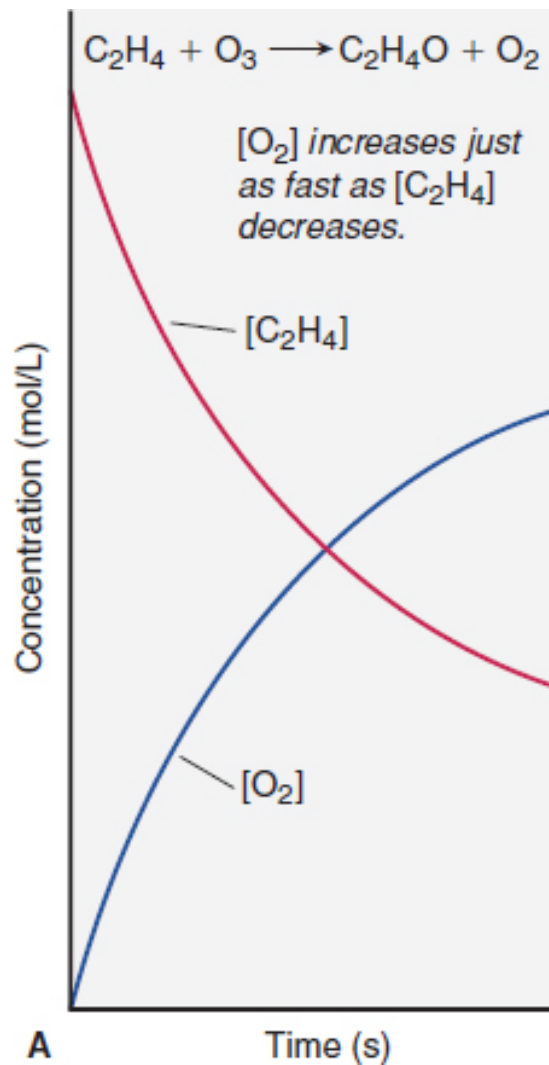
Initial Rate: The instantaneous rate when the reactants are just mixed ($t=0$) – can ignore the product concentrations

Expressing Reaction Rates

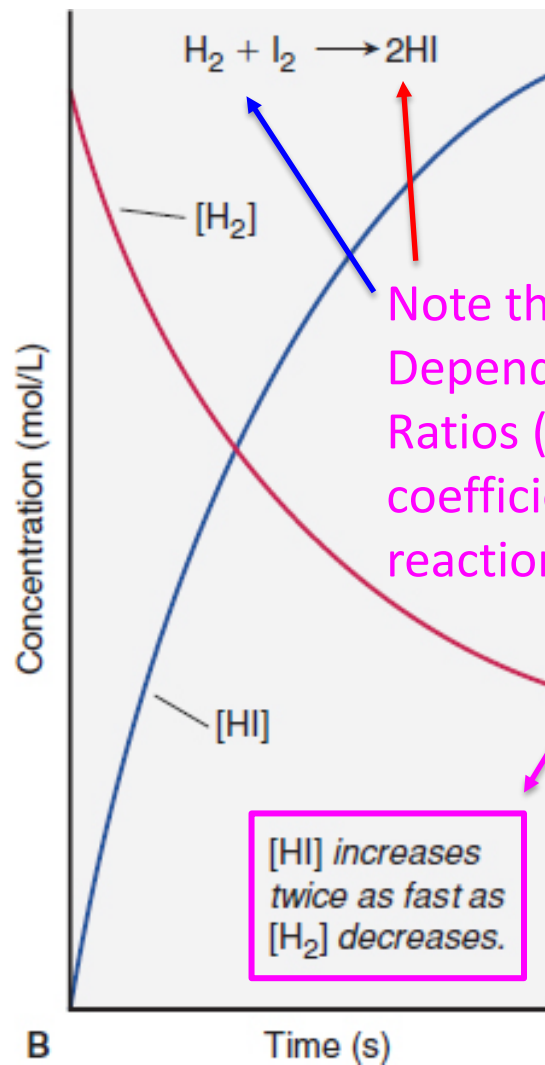


Expressing Reaction Rates

[Reactant] concentrations
Decrease with time



[Product] concentrations
increase with time



Note that the relative rates
Depend on the molar
Ratios (i.e. stoichiometric
coefficients in balanced
reaction)

[HI] increases
twice as fast as
[H₂] decreases.

Chapter 17: Kinetics

17.1-17.2: Reaction Rates

17.3: Rate Law/Reaction Orders/Rate Constant

17.4: Integrated Rate Laws

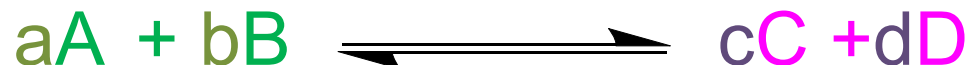
17.5: Collision and Transition State Theory

17.6: Reaction Mechanisms

17.7: Catalysis

Rate Law

Rate Law: The relationship between the rate and the concentration of products/reactants which are related by a proportionality constant, k , called **rate constant**



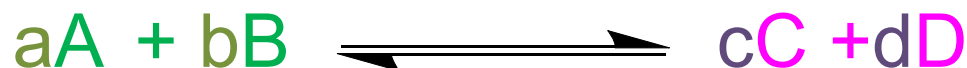
$$\text{Rate} = k [A]^m [B]^n \dots$$

where m and n are the order of the reaction in A and B, respectively

- Rate law must be determined experimentally.
- Stoichiometry **does not** necessarily predict the rate law.
- Rate law can have product and reactant terms
- The values of the exponents (m and n) can be integers, fractions, negative or positive

$$\text{Rate} = R \quad \text{Units } M s^{-1} = mol L^{-1} s^{-1}$$

Rate Law and Reaction order



$$\text{Rate} = k [A]^m[B]^n$$

where m and n are the order of the reaction in A and B, respectively

- Rate law **must be determined experimentally**.
- Stoichiometry does not necessarily predict the rate law.
- Rate law can have product and reactant terms
- The values of the exponents (m and n) can be integers, fractions, negative or positive
- Overall reaction order is the sum of exponents (m+n) in the rate law (If the sum = 0 – zero order; = 1 – first order; = 2 – second order)

What are the units of k ? units of k depend upon the order of the reaction...must yield units of $R = M s^{-1}$

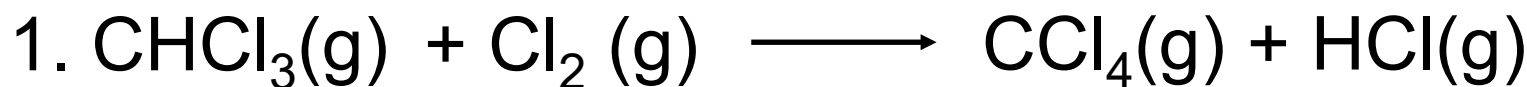
Rate Law and Reaction order

 Example Rate equations for hypothetical cases

Units of R $\text{M s}^{-1} = \text{mol L}^{-1} \text{s}^{-1}$
(always)

	Overall Reaction Order	Units of k
$R = k[A]^0 = k$	0	M s^{-1}
$R = k[A]$	1 st	s^{-1}
$R = k[A]^2$	2 nd	$\text{M}^{-1} \text{s}^{-1}$
$R = k[A][B]$	2 nd	$\text{M}^{-1} \text{s}^{-1}$
$R = k[A][B][C]$	3 rd	$\text{M}^{-2} \text{s}^{-1}$
$R = k[A]^2[B][C]^{-1}$	2 nd	$\text{M}^{-1} \text{s}^{-1}$

Determine the order of the following reactions



$$\text{Rate} = k[\text{CHCl}_3][\text{Cl}_2]^{1/2} \quad \text{Units of } k \quad \text{M}^{-\frac{1}{2}} \text{s}^{-1}$$

Order = $1 + \frac{1}{2} = 1\frac{1}{2}$ 1.5 order overall



$$\text{Rate} = k[\text{O}_3]^2[\text{O}_2]^{-1}$$

Order $\approx 2 + (-1) \approx 1$ 1st order overall

Note does not match stoichiometry...the reason will be explained in Reaction Mechanism